

Reference excerpt 02-027

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parameter (see Eq. (1)). Once identified the waves a_i , c_i , x_i , v_i and y_i on each cardiac cycle, the intervals t_{axi} and t_{vyi} are calculated for each cardiac cycle. The parameters defined in Eq. 1 are used to define the two quantities $a_i = a_i - x_i$ and $v_i = v_i - y_i$ (2). The average and the standard deviation for each parameter a_i , v_i , t_{axi} and t_{vyi} are calculated over several cardiac cycles. An illustrative JVP trace is presented in Fig.3 where the waves a , c , x , v and y are visible. The parameters a_i , x_i , v_i and y_i for each cardiac cycle i were identified by the algorithm. However, about 10 % of the points are normally not properly detected by the algorithm, while it was possible to identify them manually (see Fig. 4). The parameters a_i , v_i , t_{axi} and t_{vyi} were calculated as in Eq. (2). The mean values and standard deviations of the parameters are presented in Table 1. JVP and ECG time relationship The time interval between the wave a and the events P and R and the wave x and the event T are calculated for each cardiac cycle. These intervals are defined as: $t_{aRi} = t_{ai} - t_{Ri}$, $t_{aPi} = t_{ai} - t_{Pi}$, $t_{xTi} = t_{xi} - t_{Ti}$ (3). Table 1: Example of mean value and standard deviation of main JVP parameters are presented. a x v $a - x$ $v - y$ t_{ax} t_{vy} HR (cm²) (cm²) (cm²) (cm²) (s) (s) Bpm
1.57 0.18 1.29 0.16 1.33 0.18